Improving Crop Disease Detection Using Multi-Dataset Training and Domain Generalization

Deliverable 1

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Abstract

Crop diseases remain a major threat to agricultural productivity, farmer income, and food security. Recent deep learning methods have shown strong performance for plant disease recognition, but many models are trained on controlled image collections and fail to generalize to real-world settings. This project is based on the paper *Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach*, which proposes combining PlantDoc with a web-sourced dataset to improve robustness under diverse imaging conditions. This report presents the project background, selected paper analysis, dataset plan, architecture explanation, baseline models, data cleaning and preprocessing pipeline, exploratory data analysis (EDA), and repository structure. The implementation follows the paper's multi-dataset design and focuses on domain shift between controlled and realistic image environments.

1 Background

Agriculture is one of the most important sectors in the global economy because it supports food production, employment, trade, and rural livelihoods. Stable agricultural output is directly related to food availability and indirectly influences economic development, poverty reduction, and social welfare. As the world population continues to grow, the demand for reliable crop production also increases, which makes crop health monitoring a major practical concern.

Plant diseases are among the most damaging threats to agricultural productivity. They reduce both crop yield and product quality, lower market value, and increase the

need for chemical intervention. In severe cases, disease outbreaks can spread rapidly and create large-scale economic losses for farmers and agricultural supply chains. The selected paper states that plant diseases are responsible for an estimated 20–40% reduction in yields for major food and cash crops, with annual global losses valued at approximately $220 billion. These figures show that early and accurate disease detection is not only a technical problem but also an agricultural and economic necessity.

Traditionally, crop disease diagnosis has relied on manual field inspection by farmers, agricultural workers, or plant pathologists. In this process, visible symptoms such as spots, yellowing, curling, lesions, mildew, and wilting are used to identify disease conditions. Although expert-based inspection can be useful, it has several limitations. First, it is time-consuming and labour-intensive, especially on large farms. Second, it is prone to human error because multiple diseases can share visually similar symptoms. Third, it is difficult to scale in regions where trained experts are not easily available. For these reasons, conventional diagnosis methods often lead to delayed treatment and inconsistent disease identification.

Artificial intelligence has emerged as a promising solution to this problem. Machine learning and, more specifically, deep learning have enabled automated image-based crop disease recognition. In this context, convolutional neural networks (CNNs) are especially effective because they can learn discriminative visual patterns directly from raw image data. CNN-based systems can identify disease-specific textures, color changes, lesion structures, and other visual cues without requiring handcrafted feature extraction. This makes them more scalable and often more accurate than traditional rule-based image processing pipelines.

One of the most widely known benchmark datasets in this field is PlantVillage. PlantVillage is popular because it offers a large number of labeled plant leaf images and provides a clean benchmark for training and evaluation. However, PlantVillage mainly contains images captured under controlled conditions, usually with plain backgrounds, stable lighting, and centered leaf presentation. Such characteristics make the dataset useful for benchmarking but less representative of real agricultural deployment.

This mismatch between training conditions and field conditions is commonly referred to as domain shift. In crop disease detection, domain shift appears when a model is trained on clean laboratory-like images but later tested on real-world images containing cluttered backgrounds, varying lighting, blur, occlusion, shadows, and different camera angles. In such cases, a model may perform very well during validation on curated images but fail to generalize to real farm environments. This is one of the most important practical limitations in agricultural AI.

To address this issue, multi-dataset training and domain generalization are required. Multi-dataset training exposes the model to more diverse image distributions, while domain generalization aims to learn features that remain reliable across different environ-

ments. This project adopts that viewpoint by following the selected paper's methodology. Instead of relying only on a controlled benchmark dataset, the project integrates PlantDoc and a web-sourced dataset, then constructs a combined dataset for more robust training. This design is intended to reduce overfitting to a single visual domain and improve real-world crop disease classification performance.

2 Phase 1: Paper Selection and Analysis

2.1 Selected Paper

The selected paper is:

M. S. Krishna, P. Machado, R. I. Otuka, S. W. Yahaya, F. Neves dos Santos, and I. K. Ihianle, "Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach," *J*, vol. 8, no. 1, p. 4, 2025.

Direct link: https://doi.org/10.3390/j8010004

2.2 Problem Addressed

The paper addresses the challenge of plant disease detection in diverse and uncontrolled image environments. It highlights that many existing detection systems rely heavily on lab-captured images and therefore do not generalize well to realistic field conditions.

2.3 Importance of the Problem

The problem is important because disease outbreaks directly reduce yield, damage crop quality, and threaten food security. A practical disease detection system must work not only on curated benchmark images but also on real-world plant images taken under variable agricultural conditions.

2.4 Method Used

The paper combines the PlantDoc dataset with a web-sourced dataset and evaluates several state-of-the-art CNN architectures, including EfficientNet-B0, EfficientNet-B3, ResNet50, and DenseNet201. The approach also introduces augmentation techniques such as Gaussian noise to improve generalization and robustness.

2.5 Novelty

The paper's main novelty is its multi-dataset strategy. Rather than training on a single controlled benchmark, it merges PlantDoc with web-sourced images in order to increase visual diversity and improve real-world adaptability. The paper also publishes the constructed dataset to support future research.

2.6 Architecture Explanation

The proposed architecture follows a modular deep learning pipeline:

```
Input image
   -> Data cleaning and preprocessing
   -> Data augmentation
   -> CNN feature extractor
   -> Classification head
   -> Disease label prediction
```

The selected paper describes four main components:

- data preprocessing module,
- model architecture module,
- training module,
- evaluation module.

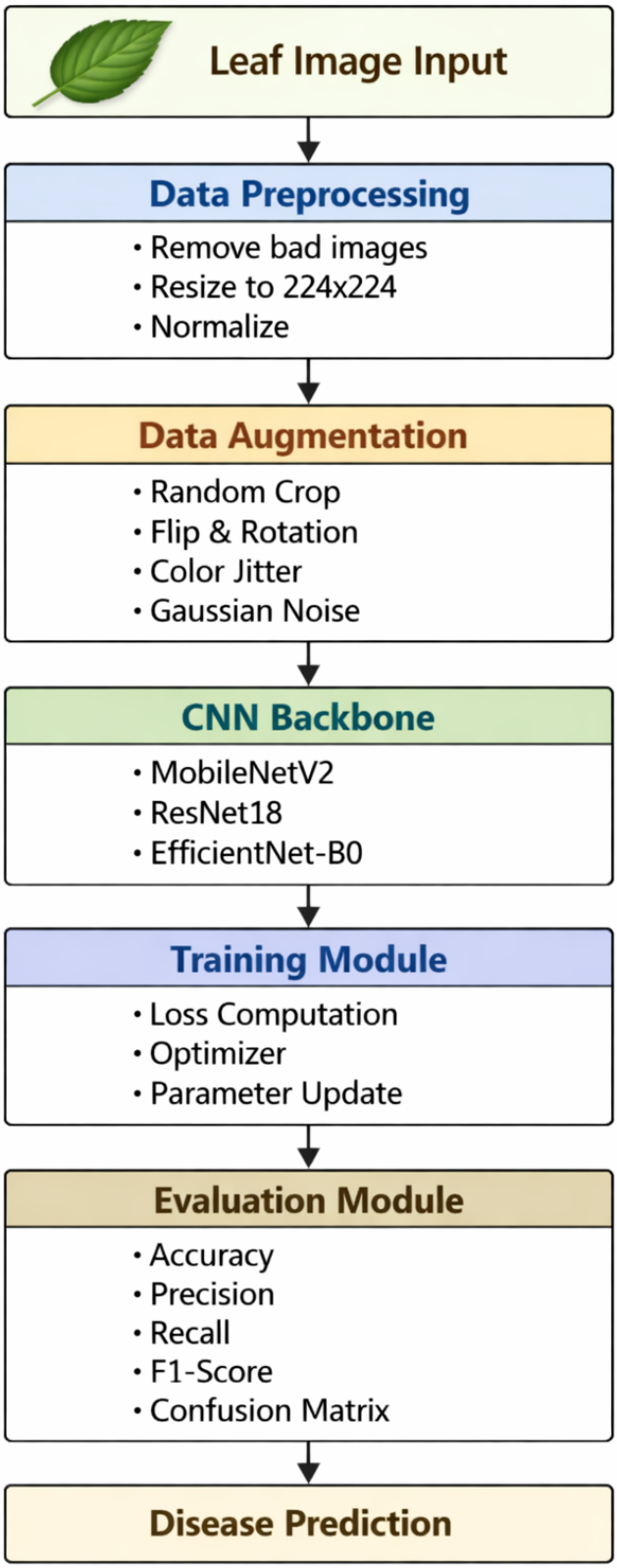

Figure 1: Overall crop disease detection pipeline based on the selected paper.

2.7 Baseline Models

For this project, two baseline models are selected for implementation:

- **MobileNetV2**: selected because it is lightweight, computationally efficient, and suitable for resource-constrained or mobile deployment scenarios.
- **ResNet18**: selected because it is a reliable CNN baseline with residual connections and moderate computational complexity.

In later stages, a stronger comparison model such as **EfficientNet-B0** will be added to align more closely with the selected paper.

3 Phase 2: Dataset Plan and Data Collection

3.1 Datasets Used

This implementation follows the paper's dataset strategy. The paper explicitly states that the primary datasets are **PlantDoc** and a **new web-sourced dataset**, and that these are merged to improve generalization under diverse plant disease scenarios.

The datasets used in this project are:

1. **PlantDoc**: used as the structured source dataset.
2. **Web-sourced dataset**: obtained from the Zenodo record provided by the paper.
3. **Combined dataset**: created by merging PlantDoc and the web-sourced dataset.

3.2 Dataset Purpose

- **PlantDoc**: provides labeled disease images and structured train/test organization.
- **Web-sourced dataset**: introduces more realistic visual diversity, including differences in background, lighting, and image quality.
- **Combined dataset**: supports more robust model learning by exposing the classifier to multiple domains.

3.3 Target Classes

Based on the class-wise experimental tables in the selected paper, the implementation focuses on the following 16 classes:

1. apple_leaf_healthy
2. apple_leaf_rust
3. apple_leaf_scab
4. corn_grey_leaf_spot

5. corn_leaf_blight
6. corn_leaf_healthy
7. corn_leaf_rust
8. potato_leaf_blight
9. potato_leafroll_virus
10. tomato_leaf_bacterial_spot
11. tomato_leaf_early_blight
12. tomato_leaf_healthy
13. tomato_leaf_late_blight
14. tomato_leaf_mould
15. tomato_leaf_powdery_mildew
16. tomato_septoria_leaf_spot

4 Data Cleaning

The project includes automated data cleaning scripts to improve dataset quality before training. The cleaning stage addresses three main issues:

- **Duplicate removal**: duplicate image files are identified through file hashing and removed to prevent repeated samples.
- **Corrupted image removal**: invalid image files are detected using image verification and deleted.
- **Noise filtering**: very small or unusable images can be removed based on minimum size thresholds.

The cleaning script is implemented in `data_scripts/cleaning.py`.

5 Preprocessing

The preprocessing pipeline standardizes all images before training. The selected paper resizes all images to **224** $\times$ **224** pixels and normalizes them using **ImageNet statistics**, which is consistent with the use of pretrained CNN backbones such as ResNet and EfficientNet.

The preprocessing stage in this project includes:

- image resizing to a fixed spatial resolution,
- pixel normalization,
- class label encoding,
- dataset splitting into training, validation, and testing subsets.

The preprocessing script is implemented in `data_scripts/preprocessing.py`, while dataset merging and splitting are implemented in `merge_datasets.py` and `split_datasets.py`.

6 Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and composition of the datasets before model training. The notebook `notebooks/eda.ipynb` was created to analyse:

- PlantDoc class distribution,
- web-sourced dataset class distribution,
- combined dataset class distribution,
- final train/validation/test distribution,
- representative sample images.

6.1 Purpose of EDA

EDA is important in this project because class imbalance, inconsistent class coverage, and dataset variation can affect CNN performance and bias learning toward overrepresented categories. Visualizing the dataset also helps validate that the combined dataset has been merged correctly.

6.2 Figure Placeholders for Generated EDA

After running `eda.ipynb`, export the generated charts to image files and place them in the report folder. Then replace the placeholders below with the actual filenames.

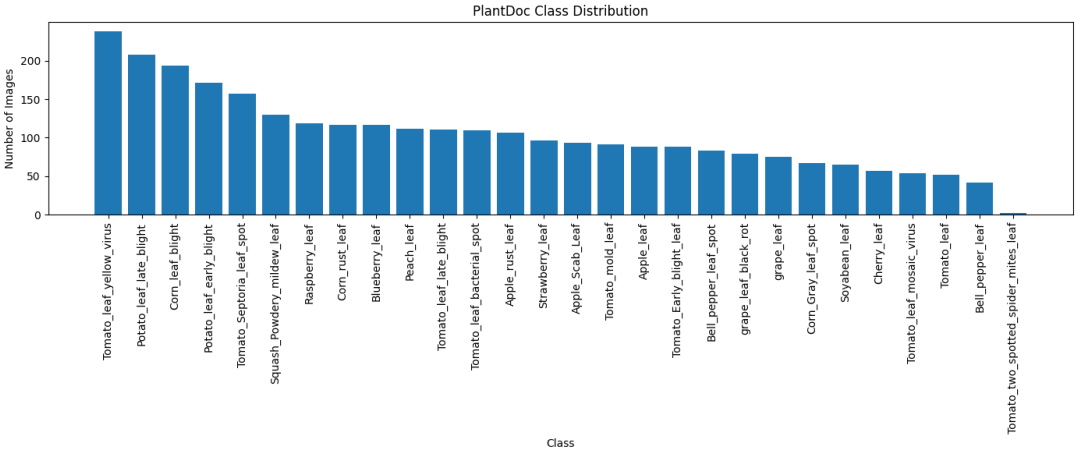

Figure 2: Class distribution of the PlantDoc dataset.

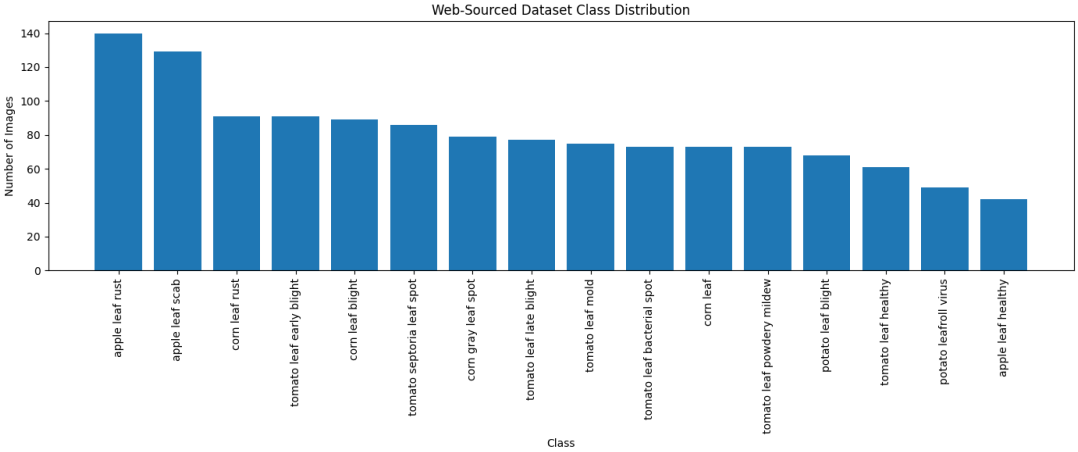

Figure 3: Class distribution of the web-sourced dataset.

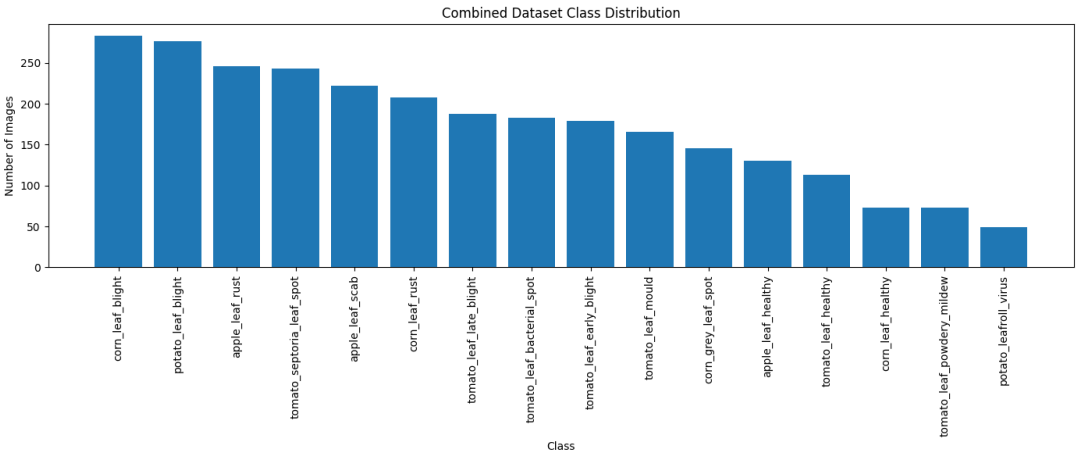

Figure 4: Class distribution of the combined dataset.

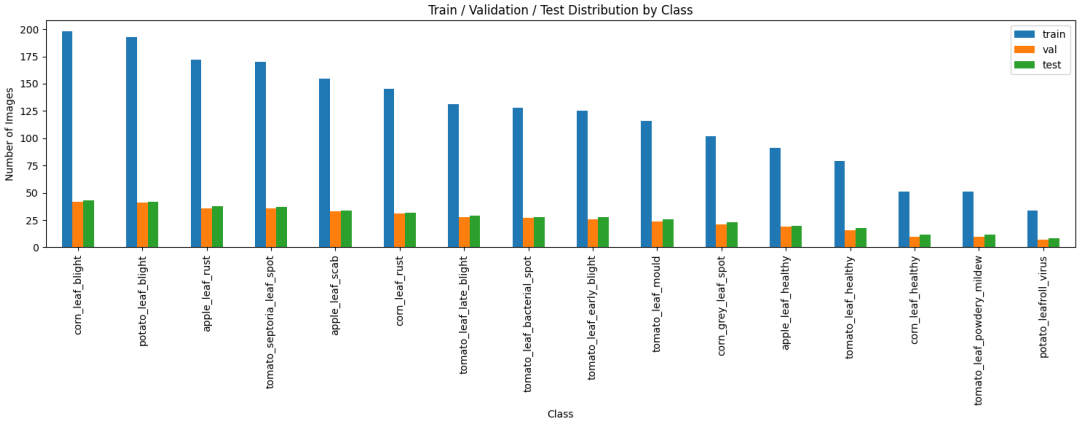

Figure 5: Final train, validation, and test distribution across the combined dataset classes.

6.3 Sample EDA Discussion

The class distribution plots reveal the relative number of samples available in each disease category and highlight the effect of combining PlantDoc with the web-sourced dataset. These visualizations are expected to show that some classes are more heavily represented than others, which may influence classification performance. The EDA also helps confirm whether the final train/validation/test split preserves approximate class balance and whether the merged dataset increases diversity compared with each source dataset individually.

7 Repository Structure

The project repository is organized as follows:

```
/project-root
 README.md
 requirements.txt
 context.txt

 data/
    plantdoc/
    web_sourced/
    combined_dataset/
    train/
    val/
    test/

 data_scripts/
```

```
    cleaning.py
    preprocessing.py
    merge_datasets.py
    split_datasets.py

models/
    baseline_train.py

notebooks/
    eda.ipynb

reports/
     deliverable1.tex
```

8 Conclusion

This deliverable establishes the academic and technical foundation of the project. The selected paper motivates the need for plant disease detection systems that can generalize across controlled and real-world image domains. Following the paper's methodology, this project uses PlantDoc and a web-sourced Zenodo dataset to construct a combined multi-dataset training pipeline. The report has documented the project background, selected paper, architecture, baseline models, dataset strategy, cleaning and preprocessing pipeline, EDA plan, and repository structure. The next phase will focus on executing the full training workflow, benchmarking baseline CNN models, and evaluating how multi-dataset training affects generalization performance.

References

1. M. S. Krishna, P. Machado, R. I. Otuka, S. W. Yahaya, F. Neves dos Santos, and I. K. Ihianle, "Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach," *J*, vol. 8, no. 1, p. 4, 2025.

2. S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science*, vol. 7, 2016.

3. E. C. Too, Y. Li, S. Njuki, and Y. Yingchun, "A comparative study of fine-tuning deep learning models for plant disease identification," *Computers and Electronics in Agriculture*, vol. 161, pp. 272–279, 2019.